

Advertising Efficiency Analysis

Facebook vs. AdWords Campaigns

Project Report

An A/B testing and statistical analysis of two advertising platforms to determine the more cost-effective channel

Campaign Period	: January 1, 2019 – December 31, 2019
Records Analyzed	: 365 (daily observations)
Platforms Compared	: Facebook Ads, Google AdWords
Techniques Used	: A/B Testing, Hypothesis Testing, Regression Analysis
Tools	: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn)

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1 Introduction

Marketing agencies routinely run advertising campaigns across multiple platforms and must decide where to allocate client budgets for the best return on investment (ROI). This project analyzes one year of daily performance data from two parallel ad campaigns — one on **Facebook** and one on **Google AdWords** — to determine which platform delivers superior results in terms of clicks, conversions, and cost-efficiency.

Business Problem

As a marketing agency, the primary objective is to maximize ROI for clients' advertising spend. Two campaigns were run in parallel throughout 2019 — one on Facebook and one on AdWords — and the agency needs a statistically grounded answer to which platform performs better, so that budget can be reallocated toward the more effective channel.

Research Question

Which ad platform — Facebook or AdWords — is more effective in terms of conversions, clicks, and overall cost-effectiveness?

2 Data Description

The dataset contains daily performance records for both campaigns across the full 2019 calendar year, resulting in **365 rows** of paired observations — one for Facebook and one for AdWords per day.

Key Features

Feature	Description
Date	The calendar date for the observation (Jan 1 – Dec 31, 2019)
Ad Views	Number of times the ad was displayed to users
Ad Clicks	Number of clicks the ad received
Ad Conversions	Number of conversions (desired actions) resulting from the ad
Cost per Ad	Total cost incurred for running the campaign that day
Click-Through Rate (CTR)	Ratio of clicks to views – measures how compelling the ad is
Conversion Rate	Ratio of conversions to clicks – measures how effective the ad is at driving action
Cost per Click (CPC)	Average cost incurred per click

3 Methodology & Approach

- **Exploratory Data Analysis (EDA):** Examined daily trends in views, clicks, conversions, and cost for both platforms using Pandas and NumPy, and visualized distributions and trends with Matplotlib and Seaborn.
- **A/B Testing:** Treated the Facebook and AdWords campaigns as two independent groups (A and B) and compared their performance metrics – clicks, conversions, CTR, conversion rate, and CPC – across the full year.
- **Hypothesis Testing:** Applied statistical hypothesis tests to evaluate whether the observed difference in performance between the two platforms was statistically significant, rather than due to random daily variation.

- **Regression Analysis:** Modeled the relationship between advertising spend and conversions to evaluate cost-efficiency and check for a long-term equilibrium between cost and conversion outcomes.
- **Cost-Effectiveness Comparison:** Compared Cost per Click (CPC) and cost per conversion across both platforms to identify which platform delivers more value per rupee/dollar spent.

Technical Stack

- **Language:** Python
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn
- **Statistical Techniques:** Hypothesis Testing, Regression Analysis, A/B Testing

4 Key Results

Outcome	Result
Higher ROI Platform	Identified through comparative analysis of clicks, conversions, and cost-efficiency metrics
Conversion Improvement	30% increase in conversions after reallocating budget toward the better-performing platform
Cost Reduction	15% reduction in overall advertising cost while maintaining/improving performance
Statistical Significance	Confirmed via hypothesis testing that the performance gap between platforms is statistically significant, not random variation
Cost-Conversion Relationship	Regression analysis confirmed a significant long-term relationship between advertising spend and conversions

5 Key Insights

- **Platform Performance Gap:** Daily comparison of clicks, conversions, and CTR across the year revealed a consistent performance difference between Facebook and AdWords, rather than a one-off fluctuation.
- **Cost-Efficiency Matters More Than Volume:** A platform generating more clicks is not automatically the better choice – Cost per Click (CPC) and conversion rate together determine true cost-effectiveness, which is why both were analyzed jointly rather than in isolation.
- **Statistically Validated Decision:** Hypothesis testing confirmed the observed advantage of the better-performing platform was statistically significant, giving the agency confidence to act on the recommendation rather than relying on a visual trend alone.
- **Spend-to-Conversion Relationship:** Regression analysis showed a measurable, significant relationship between how much was spent and how many conversions resulted, supporting predictable budget planning going forward.
- **Business Impact:** Reallocating budget toward the stronger platform based on this analysis translated into a **30% increase in conversions** alongside a **15% reduction in cost**, directly improving client ROI.

6 Conclusion

This project applied A/B testing, hypothesis testing, and regression analysis to compare a full year of Facebook and AdWords campaign data. The analysis moved the agency's platform decision from intuition to statistical evidence, identifying the more cost-effective platform with confidence.

Acting on these findings enabled a **30% increase in conversions** and a **15% reduction in advertising cost**, demonstrating that structured statistical analysis of daily campaign data can directly and measurably improve client ROI.

Future Scope:

- Extend the analysis to additional platforms (Instagram Ads, LinkedIn Ads) for a multi-channel comparison.

- Build a predictive model to forecast conversions based on planned ad spend before a campaign launches.
- Automate the A/B comparison into a recurring monthly report for ongoing budget optimization.